

SET 5

1. Design and configure IPv6 addressing for a network of 8 systems using Packet Tracer. Verify communication using ping6.
2. Assume that you are the network administrator of a bank requiring secure communication between the head office and a branch office. Configure a basic VPN tunnel scenario using Packet Tracer.
3. Using Wireshark, Capture and analyze the Packets that use RTP/VoIP for different header fields and payload of the concerned packet.
4. Design a Point-to-Point Topology using Packet Tracer connecting 2 routers via a serial link, and configure IP addressing on both ends.

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1. Design and configure IPv6 addressing for a network of 8 systems using Packet Tracer. Verify communication using ping6.

Aim: To design and configure IPv6 addressing for eight systems using Packet Tracer and to verify communication between the systems using ping6.

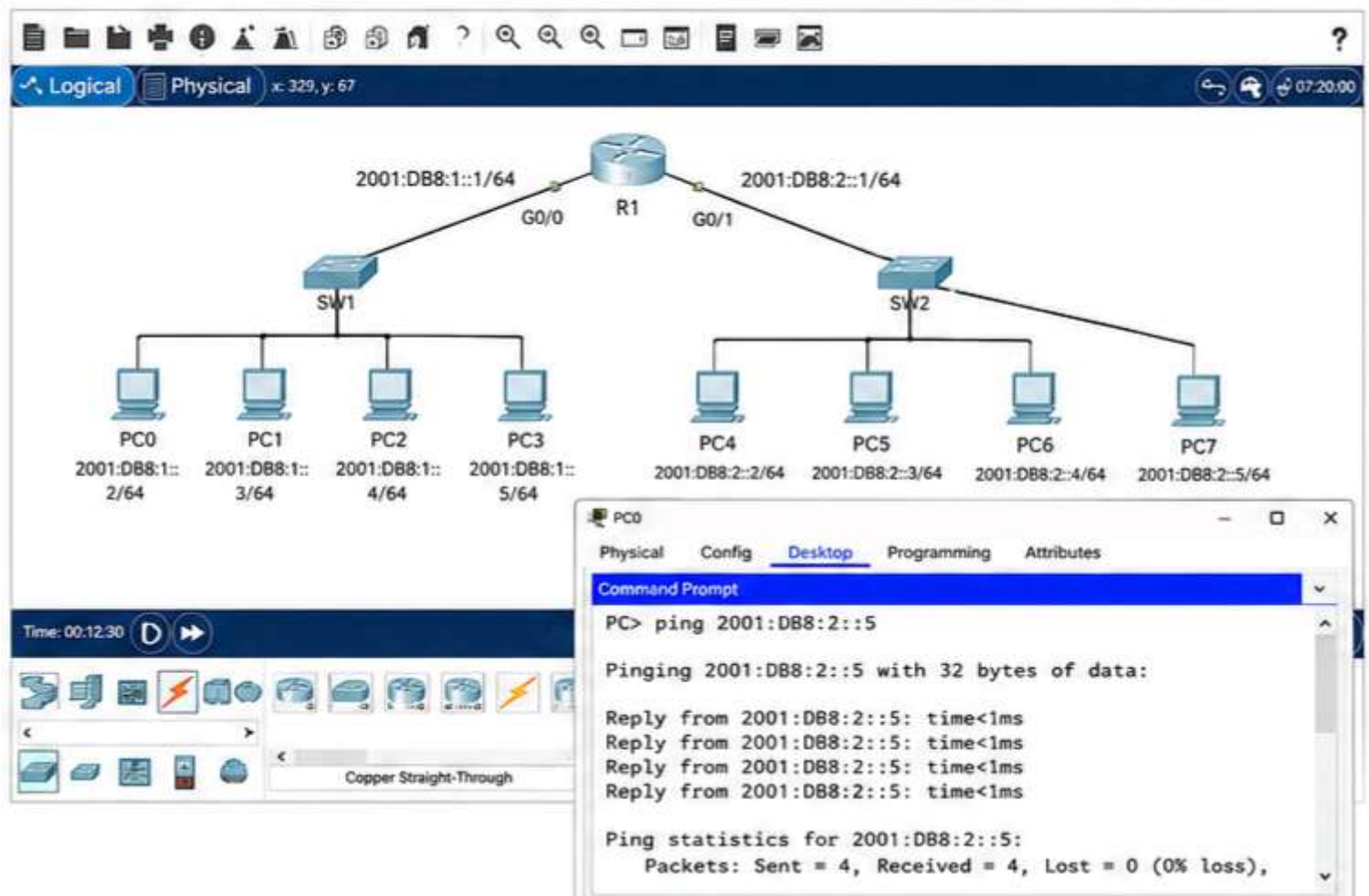
Procedure:

1. Open Cisco Packet Tracer.
2. Place one switch and eight PCs in the workspace.
3. Connect all the PCs to the switch using straight-through cables.
4. Open the IP Configuration of each PC.
5. Assign IPv6 addresses to the eight PCs using the same network.
6. The IPv6 addresses are assigned as 2001:DB8:1::1/64 to 2001:DB8:1::8/64.
7. Open the Command Prompt of any PC.
8. Use the ping6 command with the IPv6 address of another PC.
9. Check whether reply messages are received.
10. Repeat the ping test between different PCs.

Result:

Thus, IPv6 addressing was successfully configured for eight systems and communication between the systems was successfully verified using the ping command.

1. IPv6 addressing for 8 systems



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2. Configure a basic VPN tunnel between Head Office and Branch Office using Packet Tracer.

Aim:

To configure a basic VPN tunnel between the Head Office and Branch Office using Cisco Packet Tracer and to verify communication between the two networks.

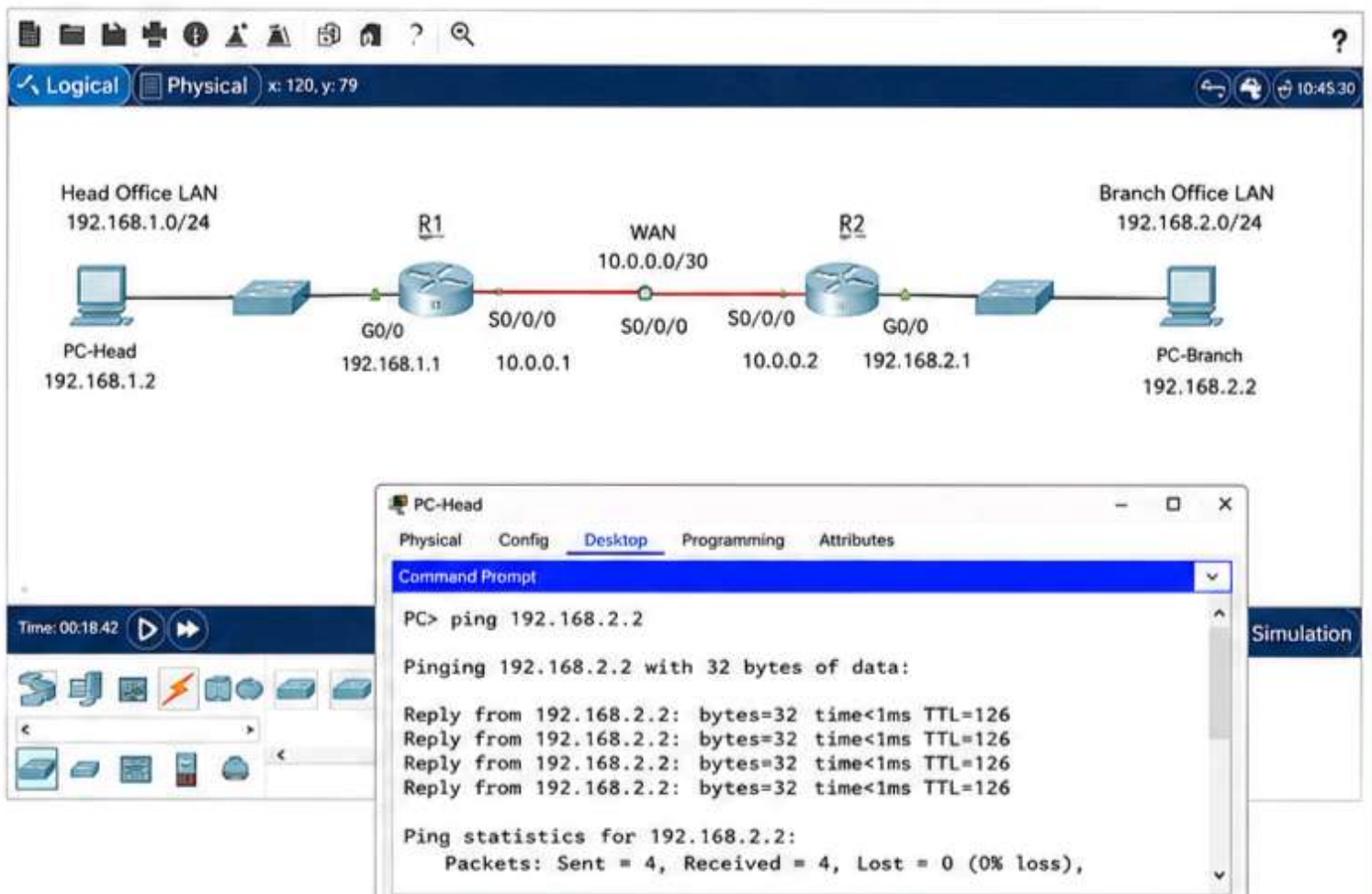
Procedure:

1. Open Cisco Packet Tracer.
2. Place two routers, two switches and PCs for the Head Office and Branch Office.
3. Connect the PCs to their respective switches.
4. Connect each switch to its router.
5. Connect the two routers using a suitable WAN link.
6. Configure IP addresses for the router interfaces and PCs.
7. Configure the routing between the Head Office and Branch Office networks.
8. Configure the required VPN tunnel parameters on both routers.
9. Enable the router interfaces using the no shutdown command.
10. Test communication between the two offices using the ping command.
11. Verify that packets are successfully transmitted between the networks.

Result:

Thus, a basic VPN tunnel was successfully configured between the Head Office and Branch Office, and communication between the two networks was successfully verified.

2. Basic VPN tunnel between Head Office and Branch Office



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3. Using Wireshark, capture and analyze packets that use RTP/VoIP for different header fields and payload.

AIM:

To capture and analyze RTP/VoIP packets using Wireshark and to study the different header fields and payload of the captured packets.

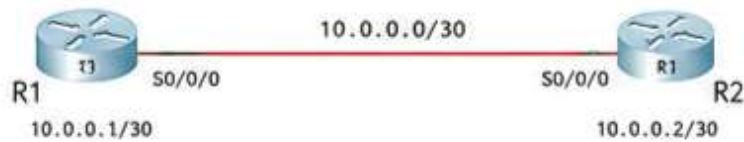
Procedure:

1. Open Wireshark on the computer.
2. Select the active network interface.
3. Start the packet capture.
4. Generate VoIP/RTP traffic through the network.
5. Allow the packets to be captured for some time.
6. Stop the packet capture.
7. Apply the filter rtp in Wireshark.
8. Select any RTP packet from the captured packets.
9. Expand the RTP section and observe its header fields.
10. Study the Version, Payload Type, Sequence Number, Timestamp and SSRC fields.
11. Observe and analyze the payload information of the packet.
12. Compare different RTP packets and note their header values.

Result:

Thus, RTP/VoIP packets were successfully captured using Wireshark and their header fields and payload were analyzed.

3. Point-to-Point topology using Serial link



```
R1# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	administratively down	
Serial0/0/0	10.0.0.1	YES	manual	up	up

```
R1# ping 10.0.0.2
```

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R1#

```
R2# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	administratively down	
Serial0/0/0	10.0.0.2	YES	manual	up	up

```
R2# ping 10.0.0.1
```

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R2#

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- 4 Design a Point-to-Point topology using Packet Tracer connecting 2 routers via a serial link, and configure IP addressing on both ends.

AIM: To design a Point-to-Point topology by connecting two routers through a serial link and to configure IP addressing on both ends using Cisco Packet Tracer.

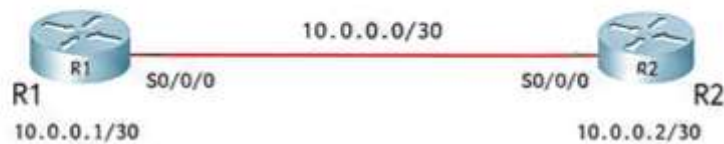
Procedure:

1. Open Cisco Packet Tracer.
2. Place two routers in the workspace.
3. Connect the two routers using a serial cable.
4. Configure the serial interface of Router 0 with IP address 10.0.0.1.
5. Configure the serial interface of Router 1 with IP address 10.0.0.2.
6. Use the subnet mask 255.255.255.252 on both interfaces.
7. Configure the clock rate on the DCE side.
8. Enable both interfaces using the no shutdown command.
9. Check the interface status using show ip interface brief.
10. Verify that both interfaces are in the up state.
11. Test the connection from Router 0 using ping 10.0.0.2.
12. Check whether successful replies are received.

Result:

Thus, a Point-to-Point topology was successfully designed using two routers connected through a serial link, and IP addressing and connectivity were successfully verified.

4. IP addressing on both ends of Serial link (Point-to-Point)



```
R1# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	administratively down	
Serial10/0/0	10.0.0.1	YES	manual	up	up

```
R1# ping 10.0.0.2
```

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R1#

```
R2# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	administratively down	
Serial10/0/0	10.0.0.2	YES	manual	up	up

```
R2# ping 10.0.0.1
```

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R2#